Saphenous Vein Ablation, Radiofrequency

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Clinical Indications

- Radiofrequency saphenous vein ablation may be indicated when **ALL** of the following are present(1)(2)(3)(4):
  - Incompetence of saphenous vein[A] documented by duplex ultrasound or other imaging test with valve closure time of greater than 500 msec(9)(10)(11)
  - Saphenous venous insufficiency symptoms causing functional impairment, including **1 or more** of the following(1)(9)(24):
    - Bleeding or ruptured superficial varicose veins
    - Leg edema
    - Leg fatigue
    - Leg pain
    - Persistent or recurrent superficial thrombophlebitis
    - Persistent or recurrent venous stasis ulcer
    - Skin changes (eg, lipodermatosclerosis, hemosiderosis)
  - No clinically significant lower extremity arterial disease(27)
  - No deep venous thrombosis on duplex ultrasound or other imaging test

Alternatives

- Alternatives include(1)(2)(4)(11):
  - Compression stockings.(28) See Graduated Compression Stockings AC for further information.
  - Laser saphenous vein ablation. See Saphenous Vein Ablation, Laser AC for further information.
  - Saphenous vein ablation with adhesive injection. See Saphenous Vein Ablation, Adhesive Injection AC for further information.
  - Saphenous vein stripping. See Saphenous Vein Stripping AC for further information.
  - Sclerotherapy. See Sclerotherapy, Leg Veins AC for further information.
  - Stab phlebectomy. See Stab Phlebectomy AC for further information.

Evidence Summary

Background

Endovenous radiofrequency ablation of the saphenous vein is a minimally invasive technique that is used to treat symptomatic saphenous vein insufficiency. Endovenous radiofrequency ablation causes thermal destruction of the venous tissues. Electrical energy is delivered through a radiofrequency catheter that has been passed percutaneously to the desired location in the saphenous vein. Ultrasonography is used to confirm and map all areas of reflux as well as to trace the path of the greater saphenous trunk from the saphenofemoral junction down the leg to the upper calf.(5)(6) **(EG 2)** Absolute contraindications to endovenous radiofrequency ablation include acute deep venous thrombosis and significant lower extremity arterial insufficiency.(2)(7)(8) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 23 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 1 Cochrane systematic review.

For saphenous vein incompetence, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society practice guideline states that a cutoff value of 500 msec confirms the diagnosis of saphenofemoral incompetence.(1) **(EG 2)** It has been estimated that radiofrequency saphenous vein ablation results in durable occlusion rates of 70% to 90%.(2)(12)(13) **(EG 2)** A prospective multicenter cohort study involving 225 patients with 295 great saphenous veins treated by radiofrequency ablation found that the initial vein occlusion rate was 100%, with a follow-up occlusion rate of 91.9% and a reflux-free rate of 94.9% at 5 years.(14) **(EG 2)** Meta-analyses and systematic reviews have concluded that endovenous radiofrequency ablation is at least as effective as conventional surgery, including high ligation and stripping, for the treatment of great saphenous vein varices.(15)(16) **(EG 1)** A meta-analysis comparing endovenous radiofrequency with alternative procedures reported that, at a mean follow-up of 32 months, the estimated pooled success rates for stripping, ultrasound-guided foam sclerotherapy, endovenous radiofrequency, and endovenous laser were 78%, 77%, 84%, and 94%, respectively. After adjusting for follow-up, endovenous radiofrequency and ultrasound-guided foam sclerotherapy were as effective as surgical stripping.(17) **(EG 1)** A systematic review and meta-analysis of 49 studies evaluating treatment of small saphenous vein incompetence, 9 of which were studies of radiofrequency ablation, found that the pooled anatomic success rate was 97.1% for radiofrequency ablation, as compared with 58.0% for conventional surgery, 63.6% for ultrasound-guided foam sclerotherapy, and 98.5% for endovenous laser ablation. The authors concluded that endovenous thermal ablation (radiofrequency or laser) should be preferred over surgery or foam sclerotherapy for the treatment of small saphenous vein incompetence.(18) **(EG 1)** A meta-analysis of 28 randomized controlled trials comparing endovenous ablation and surgical intervention in patients with varicose veins concluded that endovenous ablation (radiofrequency or laser) was associated with lower hematoma and wound infection rates, as well as less pain and quicker return to daily activities.(19) **(EG 1)** A single-center randomized trial of 450 patients with symptomatic great saphenous vein incompetence comparing endovenous laser ablation, indirect radiofrequency ablation, and direct radiofrequency ablation found, at 1-year follow-up, that both laser and indirect radiofrequency ablation were associated with improved occlusion rates compared with direct radiofrequency ablation, with no difference in occlusion rates seen between laser and indirect ablation.(20) **(EG 1)** A randomized controlled trial comparing endovenous laser ablation with radiofrequency ablation found that the techniques had comparable occlusion rates at 3 months; however, radiofrequency ablation was associated with less periprocedural discomfort.(21) **(EG 1)** Randomized trials comparing endovenous laser ablation with radiofrequency ablation in patients with varicose veins concluded that the procedures had comparable efficacy and safety at up to 5-year follow-up.(22)(23) **(EG 1)** A multi-specialty society practice guideline supports vein ablation for symptoms due to saphenous vein incompetence, including varicose veins, edema, skin changes, and ulcers.(24) **(EG 2)** Evidence-based specialty society guidelines recommend endovenous thermal ablation (radiofrequency or laser) rather than high ligation and stripping for treatment of greater saphenous vein incompetence.(3)(9)(11)(25) **(EG 2)** A consensus statement indicates that, while invasive therapy, compression therapy, and pharmacologic treatment may all offer symptomatic improvement, only invasive treatment has the potential to provide functional improvement.(26) **(EG 2)**

Alternatives

Compression therapy is sometimes used for treatment of symptomatic saphenous vein insufficiency, but patient adherence is the major factor causing failure of compression therapy. Reported rates of nonadherence have ranged from 12% to 60%, even under clinical supervision in venous ulcer clinics.(13)(29) **(EG 1)** An open-label, single-center, randomized controlled trial of 56 adults with venous leg ulcers associated with saphenous vein insufficiency compared treatment with compression alone or combined with radiofrequency ablation and found, at 12-month follow-up, that radiofrequency ablation was associated with a lower ulcer recurrence rate compared with compression therapy alone.(30) **(EG 1)** Evidence-based clinical guidelines state that patients with confirmed varicose veins and truncal reflux should first be offered endothermal ablation (radiofrequency or laser), ultrasound-guided foam sclerotherapy, or surgery prior to resorting to compression stockings for therapy.(1)(3)(9) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(31); 42 CFR 422.101(32)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(33); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(34); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(35)

Medicare Coverage Determinations: Medicare Coverage Database(36)

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Footnotes

[A] Vessels amenable to ablation include the greater saphenous vein, anterior and posterior accessory saphenous veins, and small saphenous vein.(9) [A in Context Link 1]

Codes

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